

entered, the information is gone. LSTM (long short-term memory) is a unique kind of RNN that has the capability of learning lengthy-time period dependencies. It tends to remember facts for long periods of time as their default behaviour.

LSTM units (or blocks) serve as a foundation for RNNs. An LSTM network is an RNN that is made up of LSTM units. A cell, an input gate, an output gate, and a forget gate make up a typical LSTM unit. The LSTM term 'memory' refers to the cell's responsibility for 'remembering' values over arbitrary time intervals. Each of the three gates can be called a 'traditional' artificial neuron in a multi-layer (or feed-forward) neural network; they compute the activation of a weighted sum (using an activation function). They can be considered as regulators of the flow of values that pass through the LSTM's connections, hence the name 'gate'. Between these gates and the cell, there exist connections.

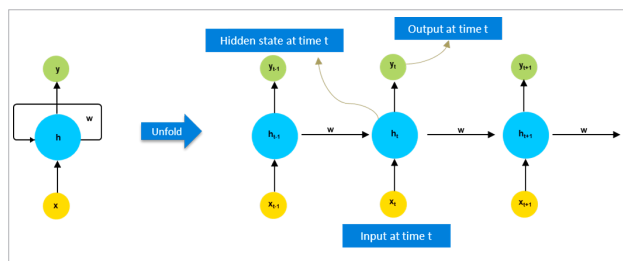


Figure 3: A Typical RNN workflow

We can understand why LSTM is preferable over RNN in the context of natural language processing and suggesting text auto-complete across longer sentences. The impact of the word early in the sentence can determine the meaning and semantics of the end of the sentence. Unlike RNN, the LSTM architecture introduces 'cell states', which can bring meaning from the beginning of the sentence to the end of the sentence. What's fascinating is that it can also be bidirectional, so the later words can provide context to the early words. Based on what word appeared at the beginning, your auto complete sentence might be different. In the example given in Figure 4, the first word in the second sentence differs based on the very first word (far away from the second sentence).

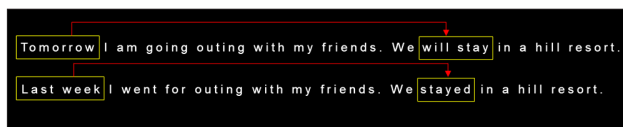


Figure 4: Autocomplete uses LSTM

Most state-of-the-art NLP models were based on RNN before the release of the transformer. RNN accesses the cell of the final word by processing data

word by word. Long sequences are difficult for RNN to handle. The model tends to forget distant position contents or, in some situations, to mix neighbouring position contents. The more steps, the more difficult it is for the recurrent network to make decisions.

LSTM outperforms traditional RNNs by a small margin. The gate mechanism is used by LSTM to determine which information the cell should remember and which it should discard. It can also solve the problem of disappearing gradients that RNN has. LSTM is adequate, but not sufficient. LSTM, like the RNN, cannot be trained in parallel. If we simplify how LSTMs function, we can say that the process is broken down into multiple steps. To begin with, LSTMs forget irrelevant information about the current state of the cell. Then they update it with new information before calculating the result.

Convolutional neural networks (CNNs) have an advantage over RNNs (and LSTMs) as they are easy to parallelise. CNNs are widely used in NLP because they are easy to train and work well with shorter texts. They capture interdependence among all the possible combinations of words. However, in long sentences, capturing the dependencies among different combinations of words can be cumbersome and unpractical

## Transformers in NLP

For decades, AI researchers have drawn their inspiration from the human brain. One of AI's most elusive ambitions is to recreate the brain's cognitive skills in deep neural networks. Attention is one of those mysterious human brain skills that we don't fully comprehend. What brain mechanisms enable us to concentrate on a single task while ignoring the rest of our surroundings? In deep learning models, attention mechanisms have recently become a source of inspiration.

A transformer, one of the biggest breakthroughs in NLP, is a deep learning model that uses the attention mechanism to weigh the importance of each element of the input data differently. The transformer model in NLP has fundamentally altered how we work with text data. The paper 'Attention Is All You Need' published in 2017, describes transformers and what is called a sequence-to-sequence architecture. Transformer is a special type of sequence-to-sequence model used for language modelling, machine translation, picture captioning and text generation. Sequence-to-Sequence (or Seq2Seq) is a neural network that converts one sequence of components into another, such as the